

PROJECT:

LAUNCH 2 EC2 INSTANCES

OS: UBUNTU, T2MEDIUM, RAM 10GB

ALLOW: SSH, HTTP, HTTPS,

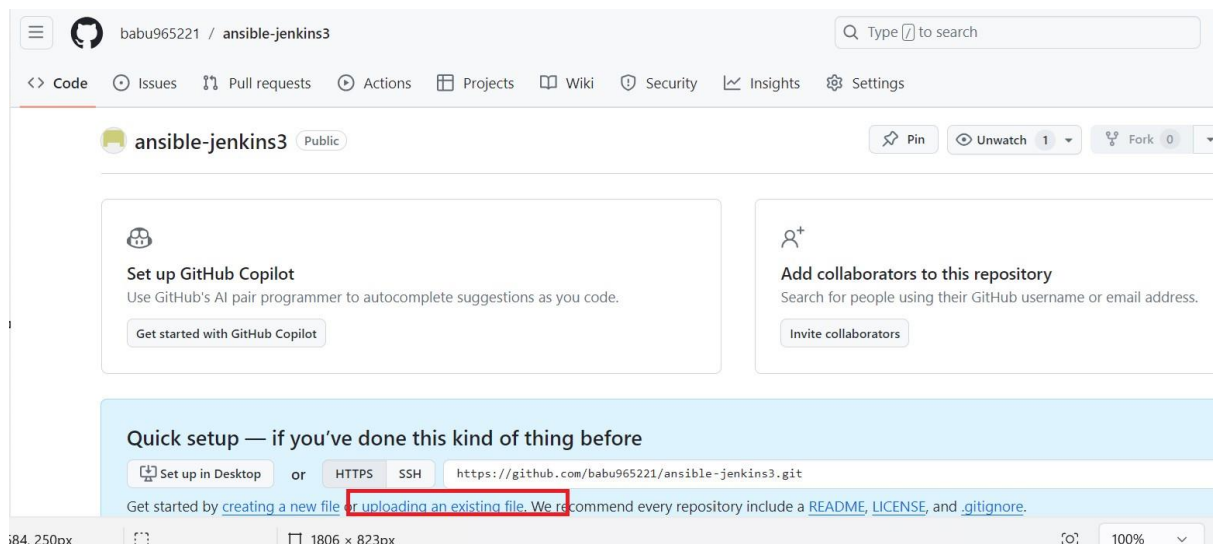
THEN LAUNCH THE INSTANCES

NAME THE INSTANCES AS 1. JENKINS 2.DOCKER

GO TO GOOGLE SEARCH BAR AND SEARCH FOR FREE CSS TEMPLATES

<https://www.free-css.com/free-css-templates>

- DOWNLOAD ANY ONE CSS TEMPLATE AND EXTRACT THE FILES
- GO TO GIT HUB AND CREATE A NEW REPO



- SELECT UPLOADING AN EXISTING FILE
- DRAG YOUR FILES TO REPO
- COMMIT THE CHANGES
- CREATE A NEW FILE DOCKER IN GIT HUB

VeeraShivaKarthik Update Dockerfile		a520893 · 4 days ago	🕒 27 Commits
📁 Sbs Free Website Template - Free-CSS.com/sbs...	Add files via upload	last week	
📄 Dockerfile	Update Dockerfile	4 days ago	

 VeeraShivaKarthik Update Dockerfile

Code Blame 2 lines (2 loc) · 47 Bytes  Code 55% faster with GitHub Copilot

```
1 FROM nginx:latest
2 COPY . /usr/share/nginx/html
```

- COMMIT THE CHANGES
- GO TO CONSOLE START YOUR INSTANCES USING PUTTY
- INSTALL THE JENKINS AND JAVA ON JENKINS SERVER(SERVER1)
- <https://www.jenkins.io/doc/book/installing/linux/#debianubuntu>

COPY THIS AND INSTALL JENKINS

- `sudo wget -O /usr/share/keyrings/jenkins-keyring.asc \`
- `https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key`
- `echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc]"`
- `https://pkg.jenkins.io/debian-stable binary/ | sudo tee \`
- `/etc/apt/sources.list.d/jenkins.list > /dev/null`
- `sudo apt-get update`
- `sudo apt-get install Jenkins`

COPY THIS AND INSTALL JAVA

- `sudo apt update`
- `sudo apt install fontconfig openjdk-17-jre`
- `java -version`
- `sudo snap install openjdk`

Start Jenkins

You can enable the Jenkins service to start at boot with the command:

```
sudo systemctl enable jenkins
```

You can start the Jenkins service with the command:

```
sudo systemctl start jenkins
```

You can check the status of the Jenkins service using the command:

```
sudo systemctl status jenkins
```

Inbound rules Info

Security group rule ID	Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
sgr-077d8d5a3bc715b04	HTTPS	TCP	443	Custom	Q 0.0.0.0/0 X	Delete
sgr-04f15adda241fbb25	SSH	TCP	22	Custom	Q 0.0.0.0/0 X	Delete
sgr-0c2514b76380bce8c	HTTP	TCP	80	Custom	Q 0.0.0.0/0 X	Delete
-	Custom TCP	TCP	8080	Anywh...	Q 0.0.0.0/0 X	Delete

Add rule

- IN IN BOUND RULES ADD NEW RULE ALLOW PORT 8080
- START YOUR JENKINS
- INTALL ANSIBLE IN JENKINS SERVER(SERVER1)
- https://docs.ansible.com/ansible/latest/installation_guide/installation_distros.html#installing-ansible-on-ubuntu
- \$ sudo apt update
- \$ sudo apt install software-properties-common
- \$ sudo add-apt-repository --yes --update ppa:ansible/ansible
- \$ sudo apt install ansible
- ubuntu @ip-172-31-16-192:~\$ sudo hostnamectl set-hostname Jenkins
- ubuntu @ip-172-31-16-192:~\$ /bin/bash
- ubuntu @jenkins:~\$
- sudo apt install python3-pip
- pip install docker --break-system-packages
- cd /etc/ansible
- ls
- sudo vim hosts

```
[dockerservers]
98.83.133.27

# This is the default ansible 'hosts' file.
#
# It should live in /etc/ansible/hosts
#
# - Comments begin with the '#' character
# - Blank lines are ignored
# - Groups of hosts are delimited by [header] elements
# - You can enter hostnames or ip addresses
# - A hostname/ip can be a member of multiple groups
#
# Ex 1: Ungrouped hosts, specify before any group headers:
## green.example.com
## blue.example.com
## 192.168.100.1
## 192.168.100.10
#
# Ex 2: A collection of hosts belonging to the 'webservers' group:
## [webservers]
```

- Save the changes wq!

```
ubuntu@ip-172-31-20-28:/etc/ansible$ cd ~
ubuntu@ip-172-31-20-28:~$ sudo hostnamectl set-hostname jenkins
ubuntu@ip-172-31-20-28:~$ /bin/bash
ubuntu@jenkins:~$ sudo su
root@jenkins:/home/ubuntu# su jenkins
jenkins@jenkins:/home/ubuntu$ cd ~
jenkins@jenkins:~$
```

- Mkdir playbooks
- cd playbooks/
- ssh-keygen

```
jenkins@jenkins:~/playbooks$ ls
jenkins@jenkins:~/playbooks$ cd ..
jenkins@jenkins:~$ cd .ssh
jenkins@jenkins:~/.ssh$ ls
id_ed25519 id_ed25519.pub
jenkins@jenkins:~/.ssh$ cd id_ed25519.pub
bash: cd: id_ed25519.pub: Not a directory
jenkins@jenkins:~/.ssh$ cat id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAICPiLLaY0ARZrANig6Av5d0Dvt/Pxx0jK3WmRv6xLGWe jenkins@jenkins
```

- Generating public/private ed25519 key pair.
- Enter file in which to save the key (/var/lib/jenkins/.ssh/id_ed25519):
- Created directory '/var/lib/jenkins/.ssh'.
- Enter passphrase (empty for no passphrase):
- Enter same passphrase again:
- Your identification has been saved in /var/lib/jenkins/.ssh/id_ed25519
- Your public key has been saved in /var/lib/jenkins/.ssh/id_ed25519.pub

The key fingerprint is:

SHA256:k0HW1XrroM1005ltHm1xfnQwP9DQYBIqS4YRI2yFn5M jenkins@jenkins

The key's randomart image is:

++[ED25519 256]--+

```
| ..o=. o. +o== |
| +. +o o o.+o |
| ...o+.. ..+ |
|   Eo oo . .o=|
|   ..S  . =X|
|   . o ++O|
|   = + =o|
|   . o . .|
|   |
```

+----[SHA256]-----+

- **GO TO SERVER 2 AND CONNECT AND INSTALL DOCKER**

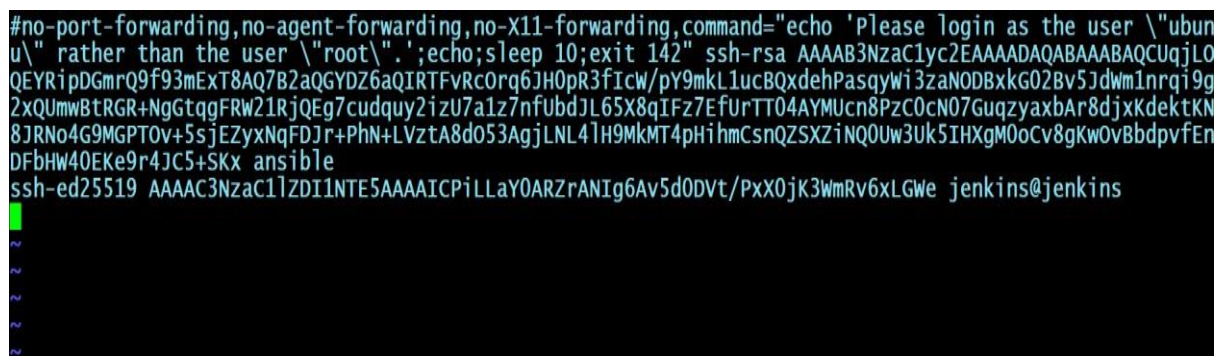
Add Docker's official GPG key:

- sudo apt-get update sudo apt-get install ca-certificates curl
- sudo install -m 0755 -d /etc/apt/keyrings
- sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
- sudo chmod a+r /etc/apt/keyrings/docker.asc

Add the repository to Apt sources:

```
echo \  
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc]  
https://download.docker.com/linux/ubuntu \  
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \  
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null  
sudo apt-get update
```

- sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
- docker ps
- we get the error so called permissions denied
- sudo usermod -aG docker ubuntu
- newgrp docker
- docker ps
- sudo su
- ssh-keygen
- cd ~
- ls -la
- cd .ssh
- ls
- vim authorized-keys(need to comment and copy the ssh keyof jenkins server to docker server)



```
#no-port-forwarding,no-agent-forwarding,no-X11-forwarding,command="echo 'Please login as the user \"ubuntu\" rather than the user \"root\".';echo;sleep 10;exit 142" ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCUqjLQ  
QEYRipDGmrQ9f93mExT8AQ7B2aQGYDZ6aQIRTFvRcOrq6JH0pR3fIcw/pY9mKL1ucBQxdehPasqywi3zaNODBxkG02Bv5Jdwm1nrqi9g  
2xQumwBTRGR+NgGtqgFRw21RjQEg7cudquy2izU7a1z7nfUbdJL65X8qIFz7EfUrTT04AYMUcn8PzC0cN07GuqzyaxbAr8djkKdektKN  
8JRNo4G9MGPTov+5sjEZyxNqFDJr+PhN+LVztA8d053AgjLNL41H9MkMT4pHihmCsnQZSXZiNQ0Uw3Uk5IHxgM0oCv8gKwOvBbdpvfEn  
DFbHW40Eke9r4JC5+SKx ansible  
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAICPiLLay0ARZrANig6Av5d0Dvt/Pxx0jK3WmRv6xLGwe jenkins@jenkins
```

- sudo systemctl reload ssh
- go to Jenkins server and connect docker server from here
- ssh root@publicip of docker
- exit
- ls
- cd ..
- cd playbooks/
- vim deployment.yaml

```
ubuntu@jenkins: /var/lib/jenkins/playbooks
--
name: Build & Deploy Docker Container
hosts: dockerservers
gather_facts: false
remote_user: root
tasks:
  - name: Copy the files to remote server
    synchronize:
      src: /var/lib/jenkins/workspace/ansible-jenkins-pipeline/
      dest: /root/project/
      recursive: yes # Ensure the entire directory is copied

  - name: Build Docker Image
    docker_image:
      name: devops-project-1
      source: build
      build:
        path: /root/project # Ensure the correct path to the Dockerfile is used
      state: present
      docker_host: unix:///var/run/docker.sock

  - name: Creating the Container
    docker_container:
      name: devops-project-1
      image: devops-project-1:latest
      ports:
        - "80:80"
```

- Save the file wq!
- Go to Jenkins server
- Create a freestyle project (ansible-jenkins-pipeline)
- Go to Source Code Management

Configure

- General
- Source Code Management**
- Build Triggers
- Build Environment
- Build Steps
- Post-build Actions

Source Code Management

☐ None

☒ Git ?

Repositories ?

Repository URL ?

https://github.com/VeeraShivaKarthik/Devops-Project-1.git

Credentials ?

VeeraShivaKarthik/*****

+ Add

Advanced ▾

Add Repository

Branches to build ?

Save Apply

- Build Now
- After your job successful go to git hub and add webhook

VeeraShivaKarthik / Devops-Project-1

<> Code Issues Pull requests Actions Projects Wiki Security 1 Insights Settings

General

Access

Collaborators

Moderation options

Code and automation

Branches

Tags

Rules

Actions

Webhooks

Environments

Codespaces

Pages

Security

Code security

Deploy keys

Webhooks / Manage webhook

Settings Recent Deliveries

We'll send a POST request to the URL below with details of any subscribed events. You can also specify which data format you'd like to receive (JSON, x-www-form-urlencoded, etc). More information can be found in [our developer documentation](#).

Payload URL *

http://54.198.8.183:8080/github-webhook/

Content type *

application/x-www-form-urlencoded

Secret

SSL verification

By default, we verify SSL certificates when delivering payloads.

☒ Enable SSL verification ☐ Disable (not recommended)

- Copy the Jenkins url and paste in webhook as shown in above image
- Select Let me select individual events.
- Select pull request and pushes

☐ Pull request reviews

Pull request review submitted, edited, or dismissed.

☒ Pull requests

Pull request assigned, auto merge disabled, auto merge enabled, closed, converted to draft, demilestoned, dequeued, edited, enqueued, labeled, locked, milestone, opened, ready for review, reopened, review request removed, review requested, synchronized, unassigned, unlabeled, or unlocked.

☒ Pushes

Git push to a repository.

☐ Registry packages

Registry package published or updated in a repository.

- Add webhook
- In git hub go to your reppo codes open your docker file and make some changes and commit the changes
- Then it automatically build the job in Jenkins because of webhook configuration
- Got Jenkins server terminal
- jenkins@jenkins:~/playbooks\$ scp -r /var/lib/jenkins/workspace/ansible-jenkins-pipeline/ [root@54.161.91.49:/root/project/](#)
- go to docker server
- now go and check the docker server

- ls
- cd ansible-jenkins-pipeline
- ls
- mv ./* ../
- cd ..
- ls
- for ref please check bellow image

```

Last login: Mon Dec  2 04:56:47 2024 from 54.172.161.74
root@ip-172-31-47-16:~# ls
project  snap
root@ip-172-31-47-16:~# cd project
root@ip-172-31-47-16:~/project# ls
Dockerfile  'Sbs Free Website Template - Free-CSS.com'  ansible-jenkins-pipeline
root@ip-172-31-47-16:~/project# cd ansible-jenkins-pipeline
root@ip-172-31-47-16:~/project/ansible-jenkins-pipeline# ls
Dockerfile  'Sbs Free Website Template - Free-CSS.com'
root@ip-172-31-47-16:~/project/ansible-jenkins-pipeline# pwd
/root/project/ansible-jenkins-pipeline
root@ip-172-31-47-16:~/project/ansible-jenkins-pipeline#

```

Go to Jenkins server terminal

- Run the command ansible-playbook deployment.yaml
- Go to Jenkins portal
- Go job configuration add buildstep (exicuteshell)

Configure

- General
- Source Code Management
- Build Triggers
- Build Environment
- Build Steps**
- Post-build Actions

Build Steps

≡ Execute shell ?

Command

[See the list of available environment variables](#)

```

scp -r /var/lib/jenkins/workspace/ansible-jenkins-pipeline/ root@54.161.91.49:/root/project/
ansible-playbook -i /var/lib/jenkins/playbooks/ansible_inventory /var/lib/jenkins/playbooks/deployment.yaml

```

- Bulid now ,Then check the console output
- Now copy the public ip of docker (target server) and paste it in web check your output.
- If You get the nginx website then you have to follow these steps.
- Go to the Docker terminal enter the command docker ps the container should be created.
- Type this command in the docker terminal docker exec -it <container_name_or_id> bash. you will enter into the container.
- Vim /usr/share/nginx/html

- If you see the default index.html file there, it means the container is using the default Nginx configuration. To serve your own files, you'll need to replace this content with your application files.
- Replace Default Nginx Content with Your App.
- If you're intending to deploy a specific app inside the container, you'll need to:
- Copy your application's files into the container and place them in the directory Nginx is configured to serve from (e.g., /usr/share/nginx/html).
- Update the Nginx configuration (if needed) to point to the correct location of your application files.
- You can copy your application files to the container using docker cp:
- `docker cp /path/to/custom/nginx.conf docker image name:/etc/nginx/nginx.conf.`
- Restart Nginx Inside the Container.
- Before copying execute this step `vim /etc/nginx/nginx.conf`.
- You should add server block.
- And, Save the file wq!.
- Restart the nginx, copy the public ip of docker and paste it in the browser
- Your website or Application will be open successfully.
- The Below Image is the Output of my web Application.

